INCH-POUND
MIL-PRF-3098/24F
05 March 2010
SUPERSEDING
MIL-PRF-3098/24E
9 December 2004

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR45/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 455 kHz; fundamental; noncontrolled; series resonance.

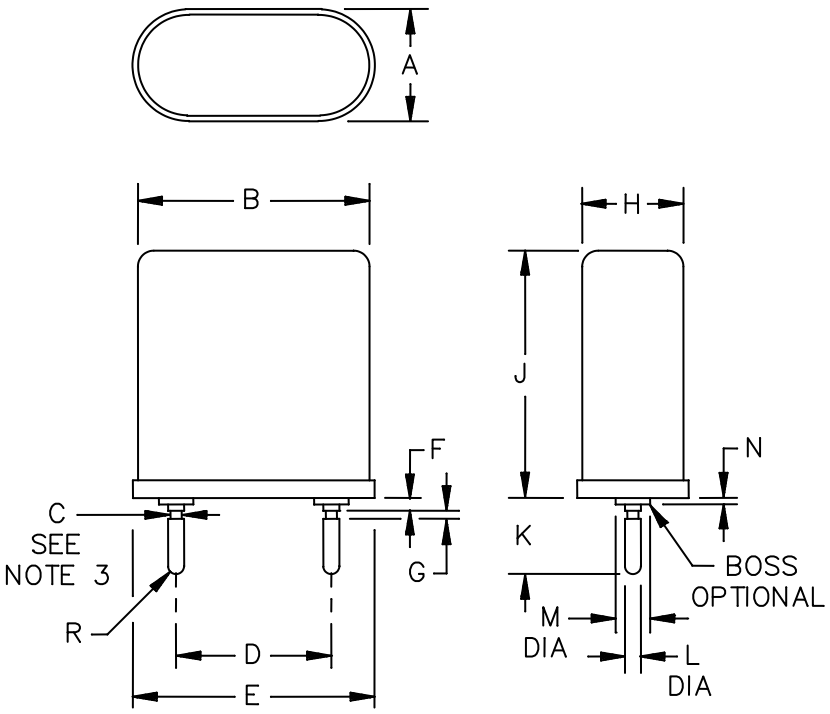



FIGURE 1. Crystal unit - CR45/U.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:22Z
Check the source to verify that this is the current version before use.

| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .352 | --- | 8.94 |
| B | --- | .725 | --- | 18.42 |
| C | .030 | .037 | 0.76 | 0.94 |
| D | .478 | .494 | 12.14 | 12.55 |
| E | --- | .757 | --- | 19.23 |
| F | .030 | .040 | 0.76 | 1.02 |
| G | .015 | .025 | 0.38 | 0.64 |
| H | --- | .317 | --- | 8.05 |
| J | --- | .775 | --- | 19.69 |
| K | .223 | .248 | 5.66 | 6.30 |
| L | .048 | .052 | 1.22 | 1.32 |
| M | .075 | .141 | 1.91 | 3.58 |
| N | .015 | .025 | 0.38 | 0.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR45/U - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:22Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 455 kHz.

Capacitance, shunt: 5.0 pF ±2.5 pF.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Calibration values:

Equivalent resistance: 2,300 ohms.

Resistor voltage drop: 2.14 volts.

Operating temperature range (noncontrolled): -40°C to +70°C, inclusive.

Frequency tolerance: ±200 parts per million (ppm). There shall be a difference in frequency of 80 Hz ±12 Hz between operation at series resonance and operation at antiresonance into load capacitance of 32.0 pF ±0.5 pF. Both measurements for this purpose shall be made at room temperature.

Equivalent resistance: 3,300 ohms, maximum.

Shock (specified pulse):

Frequency change permitted: ± 30 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202, amplitude .015 inch (0.38 mm) (.030 inch (0.76 mm) total excursion).

Frequency change permitted: ±30 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature run:

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: 500 grams, minimum.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:22Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): CR45/U followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2009-041)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:22Z
Check the source to verify that this is the current version before use.